

N-CENTRAL CVE-2026-18577 CLOUDFLARED SERVICE

Endpoint hunt for source-confirmed
Cloudflare Tunnel service registration after N-
central compromise.

HUNTING QUERY

CONFIDENCE: MEDIUM

SOURCE-CONFIRMED BEHAVIOR

retain facts, isolate interpretation

DeviceRegistryEvents

direct telemetry boundary

HUNT · VALIDATE · TUNE

no automatic malicious verdict

REPORT DATE

03 August 2026

VERSION

v1.0-corrected

PRIMARY TELEMETRYMicrosoft Defender XDR ·
DeviceRegistryEvents**ATT&CK**

T1543.003

Evidence-led dossier. Query status remains an untested implementation sketch until
executed in the target environment.

INDEPENDENT RESEARCH FORMAT · CONTROLLED UI/UX MASTER



EXECUTIVE ASSESSMENT

Decision-grade summary for detection engineering and SOC operations.

ASSESSMENT

EXECUTIVE SUMMARY

Hunt the registered service, then verify ownership.

N-able directly identifies a new CloudFlare tunnel service and the service name Cloudflared as post-exploitation evidence. The Registry key is observable, but the same service name is legitimate by default.

OPERATIONAL VALUE

Rapid scope for an RMM control-plane compromise.

The hunt can identify managed endpoints requiring triage after a high-impact N-central pivot, provided analysts validate ImagePath, tunnel ownership and change records.

8.9

01

OPERATIONAL VALUE



Medium

02

DETECTION CONFIDENCE



Hunting

03

OPERATIONAL READINESS



SOURCE-CONFIRMED FACTS

Observed behavior

- N-central versions before 2026.3.1.7 were exploitable.
- The actor obtained remote administrative access and used Take Control.
- A new CloudFlare tunnel service was registered for persistence.

ANALYTICAL ASSESSMENT

What the evidence supports

- The service Registry key is the most direct public endpoint artifact.
- Published IPs are weak standalone evidence and may be VPN exits.
- Cloudflared must be enriched before malicious disposition.

DETECTION ASSUMPTIONS

What must be validated

- MDE captures Registry modifications on managed endpoints.
- Retention covers the incident period.
- The service kept its source-reported default name.

Confidence boundary

Cloudflare documents cloudflared as the legitimate default Windows service name. A match is not proof of exploitation, tunnel success or operator identity.

02

THREAT MECHANICS

Source-observed chain and explicit observability boundaries.

ATTACK FLOW

01

Exploit N-central

Remote administrative access to a vulnerable server.

02

Use Take Control

Built-in RMM reaches managed endpoints.

03

Register service

A CloudFlare tunnel service is created.

04

Retain access

Tunnel persists after server access is revoked.

Observable core

Registry creation or modification under HKLM\SYSTEM\CurrentControlSet\Services\cloudflared, including ImagePath when populated.

Non-observable boundary

The exploit request, N-central authentication context, Take Control authorization, tunnel token, contents and remote operator.

DETECTION HYPOTHESIS

The persistence service should leave a structured Registry event.

If the attacker registers the documented tunnel service, DeviceRegistryEvents should show the cloudflared service key or values. Ownership, ImagePath, signer/hash, approved tunnel inventory and change records distinguish legitimate deployment.

REQUIRED TELEMETRY

DeviceRegistryEvents

- MDE Registry events.
- RegistryKey and value data.
- DeviceId and timestamp.

OPTIONAL ENRICHMENT

Context that improves disposition

- N-central audit and Take Control logs.
- CMDB and Cloudflare Tunnel inventory.
- Signer, hash, file and network timeline.

UNAVAILABLE HERE

Do not infer

- Exploit request and operator identity.
- Tunnel token or encrypted contents.
- Causal Take Control link in one event.



TELEMETRY & EVIDENCE MODEL

Minimum data contract and quality gates for a defensible hunt.

DATA CONTRACT

PRIMARY DATA SOURCE

Microsoft Defender XDR · DeviceRegistryEvents

Records Registry creation and modification with device, value and available initiating-process context.

EVIDENCE BOUNDARY

A matching event is a lead, not a verdict.

The event can establish service configuration activity; it cannot establish malicious ownership or successful tunneling.

REQUIRED AND ENRICHMENT FIELD MODEL

CATEGORY	FIELD	PURPOSE	REQUIREMENT
Registry event	DeviceRegistryEvents	Primary service-key evidence.	REQUIRED
Service key	RegistryKey	Exact cloudflared location.	REQUIRED
Configuration	RegistryValueName, RegistryValueData	ImagePath and service values.	REQUIRED
Device	DeviceId, DeviceName	Endpoint correlation.	REQUIRED
Event identity	Timestamp, ReportId	Investigable raw event.	REQUIRED
Creator	InitiatingProcess*	Process context when populated.	OPTIONAL
Control plane	N-central audit logs	Take Control enrichment.	ENRICHMENT

Data quality requirements

- MDE coverage on N-central-managed endpoints.
- Raw Registry value visibility.
- Authoritative Cloudflare inventory.
- N-central audit-log preservation.

Known failure conditions

- Service renamed or removed.
- Activity predates retention.
- Registry event not collected.
- No inventory to distinguish legitimate tunnels.



DETECTION LOGIC

Primary-platform hunt with every condition traceable to evidence.

HUNTING QUERY

DETECTION OBJECTIVE

Find the source-reported Cloudflared service key.

Select Registry activity for the exact service root and its subkeys, retaining configuration and creator context.

ANALYTICAL LOGIC

Direct filters and investigable output.

Direct key filter; no ActionType assumption, threshold, IOC, join, sequence or aggregation.

KQL · DEMONSTRATIVE

```
// N-able N-central CVE-2026-18577 post-exploitation: Cloudflared service registration
// Classification: Hunting query - Untested implementation sketch
// Confidence: Medium
// Platform: Microsoft Defender XDR Advanced Hunting
// Telemetry prerequisite: Microsoft Defender for Endpoint DeviceRegistryEvents
//
// Source evidence: N-able states that the attacker registered a CloudFlare tunnel
// service on managed endpoints and advises checking for service name Cloudflared.
// Cloudflare documents cloudflared as the default legitimate Windows service name,
// so every result requires ownership and ImagePath validation before escalation.

let ServiceKey = @"HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\cloudflared";
DeviceRegistryEvents
| where RegistryKey == ServiceKey
  or RegistryKey startswith strcat(ServiceKey, "\\")
| project
  Timestamp,
  DeviceId,
  DeviceName,
  ActionType,
  RegistryKey,
  RegistryValueName,
  RegistryValueData,
  InitiatingProcessAccountUpn,
  InitiatingProcessFileName,
  InitiatingProcessCommandLine,
  InitiatingProcessFolderPath,
  InitiatingProcessSHA1,
  InitiatingProcessUniqueId,
  ReportId
| order by Timestamp desc
```

REVIEW FIELDS · BASELINE LOCALLY · VALIDATE BEFORE PRODUCTION

STATUS	THRESHOLD	JOIN / SEQUENCE	PRIMARY KEY	DISPOSITION
Untested hunting	None	None	DeviceId + event time	Manual triage

What a match means

The endpoint recorded creation or modification of the default Cloudflared service configuration.

What a match does not prove

CVE exploitation, unauthorized ownership, successful tunneling, malicious content or attribution.



TRIAGE & INVESTIGATION

Evidence collection before escalation.

SOC WORKFLOW

NUMBERED WORKFLOW

- 01

Confirm the raw event.
Verify key, value, device, time and available creator fields.
- 02

Confirm ownership.
Check CMDB, Cloudflare account, tunnel inventory and ticket.
- 03

Inspect ImagePath.
Collect executable path, arguments, signer and hash.
- 04

Build the endpoint timeline.
Review file, process and network events around registration.
- 05

Correlate the control plane.
Preserve N-central audit and Take Control activity.

Escalation gate
Escalate when the service is unowned or its ImagePath/ configuration deviates from approved deployments and aligns with unexpected RMM activity.

Primary benign overlap
Cloudflare Tunnel legitimately installs the default cloudflared Windows service. Do not suppress the service name globally or trust signer alone.

INVESTIGATION PIVOTS

Host
Coverage and recent changes.

Registry
ImagePath, Start, Type and timestamps.

Process
Creator lineage and account.

File
Binary path, signer and hash.

Network
Cloudflare connections and timing.

N-central
Version, audit and Take Control sessions.

DISPOSITION CRITERIA

OUTCOME	MINIMUM EVIDENCE	ACTION
BENIGN	Approved, owned and repeatable activity.	Document and tune narrowly.
SUSPICIOUS	Unknown activity without corroborated malicious follow-on.	Deepen endpoint review.
LIKELY MALICIOUS	Unauthorized match plus coherent file, process, network or control-plane evidence.	Escalate incident response.



FALSE POSITIVES & VALIDATION

Operational controls required before promotion.

QUALITY GATE

FALSE-POSITIVE MATRIX

SCENARIO	WHY IT OVERLAPS	TUNING ACTION
Approved Cloudflare Tunnel	Uses the same default service name.	Require known owner, expected ImagePath/configuration and change record.
Multiple tunnel instances	May use unique service names or custom paths.	Inventory every approved instance and identity.
Security or lab deployment	May intentionally create the service.	Scope to approved asset and test window.

Baseline plan

- Run for at least 30 days.
- Classify every service instance.
- Build inventory from owners and change records.
- Measure Registry field population.

Validation procedure

- Install cloudflared in a controlled host.
- Confirm ImagePath and creator visibility.
- Test correlation with N-central timestamps.
- Record path normalization and gaps.

TEST CASES

CASE	EXPECTED RESULT
Controlled cloudflared installation	Expected Registry match with key/value data.
Approved existing tunnel	Match and classify through inventory.
Unowned service key	Match and escalate for deeper review.
Renamed service	Expected miss; document coverage gap.

Hunting

Initial query

Candidate

After field validation

Production

Measured precision

Optimized

Stable feedback

Success

Complete events and stable ownership-based separation.

Rejection

Benign deployments remain indistinguishable after enrichment.

Failure

Controlled installation produces no usable Registry event.



ATT&CK, CONCLUSION & REFERENCES

Coverage statement, residual risk and document control.

CLOSURE

MITRE ATT&CK MAPPING

TECHNIQUE	NAME	COVERAGE STATEMENT
T1543.003	Create or Modify System Process: Windows Service	Directly covers registration or modification of the source-reported Windows service used for persistence.

Detection coverage summary

Strong for the exact service name and Registry location; weak for renamed services and pre-retention activity.

Residual risk

Legitimate Cloudflare deployments use the same service name, while attackers can rename the service or change the persistence mechanism.

ANALYST CONCLUSION

High-value impact hunt with a mandatory ownership check.

The source evidence and documented Defender schema support a precise retrospective hunt. It must remain Hunting until a tenant-controlled installation proves telemetry completeness and authorized deployments can be separated using durable environmental context.

REFERENCES

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DOCUMENT CONTROL

VERSION	DATE	STATUS	OWNER
v1.0-corrected	03 August 2026	Hunting	Detection Engineering

Publication control

Eight A4 pages. URLs are defanged. No customer data, credentials, HTML source or rendering assets are included in the public repository.